Registration-Free Supervised Tensor-PLS for Unaligned Multiparametric MRI

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Abstract

Lesion-level multiparametric MRI has no cross-subject voxel correspondence: a lesion occupies a different place, shape, and size in every patient. Tensor partial least squares (HOPLS, N-PLS) and registration-based radiomics nonetheless treat voxel (i, j, k) as the same anatomy across subjects, an assumption lesions violate. We recast each sample as an unaligned spatial measure of multivariate-feature voxels and introduce RAW-PLS (Registration-free And Warp-invariant tensor-PLS), a supervised, covariance-maximizing tensor-PLS estimator: it transports each sample onto a single learnable cross-subject template through entropic fused Gromov–Wasserstein, barycentrically projects to a fixed-size tensor, and reads out the transport-weighted low-rank inter-atom geometry, with template, transport, and head trained jointly to maximize $X \to Y$ covariance. Adopting the unrolled transport as the estimand yields population guarantees: exact isometric invariance, Lipschitz stability to warps, dictionary identifiability up to permutation, and $\sqrt{N}$-consistency. On two genuinely unaligned BraTS targets, a purely geometric structural biomarker ($n = 1181$) and clinical grade ($n = 368$), and a synthetic warp benchmark, RAW-PLS is the only consistently positive estimator under warp. It beats centroid-aligned grid-PLS, an anatomically registered SRI24 grid, HOPLS/N-PLS, a DeepSets encoder, a higher-capacity Set Transformer, and a lesion-cropped registered radiomics baseline at both affine and deformable SyN strength (all $p < 0.05$), so the advantage is not a rasterization artifact. The struc-

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tural margin is +0.354 ($p \approx 1.8 \times 10^{-15}$) and grade AUC 0.935 vs. 0.834. The structural target has a low predictability ceiling ($R^2 \approx 0.09$); RAW-PLS is the only positive estimator there, and the registration-free advantage is the central result.
Keywords: registration-free, fused Gromov–Wasserstein, tensor partial least squares, multiparametric MRI, optimal transport, supervised dimension reduction

1. Introduction

A whole-brain atlas makes voxel (i, j, k) mean the same anatomy in every subject, but a lesion has no such fixed place. Tumors, infarcts, and other focal lesions occupy a different location, shape, and extent in each patient, so the index of a voxel inside a lesion carries no cross-subject meaning. Multilinear regression methods that summarize an imaging tensor through shared per-mode loadings inherit this index from the data: Higher-Order PLS (HOPLS) [1] and N-PLS [2] fit one spatial loading vector that is reused for all samples, which is coherent only when the spatial mode is in correspondence. Registered radiomics pipelines make the same presumption, warping each image to a template so that a fixed feature grid can be read off identical coordinates [3, 4]. When the indexed axis has no shared meaning, the shared loadings describe nothing, and the estimator is misspecified at its core: no amount of tuning recovers a correspondence the data never had.

The natural reply is to register the images first and let alignment manufacture the missing correspondence. Registration aligns healthy anatomy to an atlas [5, 6, 7], but a lesion is not in the atlas, so aligning the surrounding brain cannot place one patient's necrotic core in correspondence with another's. We do not rest this argument on intuition. Section 6 shows empirically that granting an aligned-grid baseline progressively stronger registration, affine to the SRI24 atlas [5] and then deformable SyN warping to a common template [7], does not help it relative to simple centering on either target, and that our registration-free method also beats a strong lesion-cropped registered radiomics baseline that removes whole-brain

rasterization while keeping registration.

The supervised, registration-free wedge.. Optimal transport offers a route around the index. Each sample can be modeled as a measure and transported onto a shared reference, with fused Gromov–Wasserstein (FGW) coupling features and within-sample geometry [8, 9]. The existing instances of this idea are overwhelmingly *unsupervised reconstruction*. Unaligned-tensor CP decomposition reconstructs a single tensor with one mode in a reproducing-kernel space and carries no response [10, 11]; FGW dictionary learning in imaging recovers atoms that explain the inputs, again without a target [12]; and the canonical mechanism for learning FGW templates end-to-end was built for graph classification [13]. None optimizes an external response under a covariance criterion, so none is a supervised tensor-PLS estimand. That response-identifying object is the gap.

Two adjacent lines of work already exist, and we claim neither. Registration-free supervised optimal transport for mpMRI already exists: CMOT-Fusion [14] performs *within-subject cross-modal* entropic transport with a learned cost and a barycentric projection for supervised prostate-cancer risk. End-to-end learnable FGW templates also already exist, in the graph setting [13]. We claim neither adjacency. Our novelty is the *conjunction*: a supervised, covariance-maximizing tensor-PLS estimator for unaligned spatial measures, built on a learnable *cross-subject Gromov* template, with identifiability and consistency guarantees, which we call RAW-PLS (Registration-free And Warp-invariant tensor-PLS). No single conjunct is a first; the conjunction has no prior occupant.

Contributions..

(1) A registration-free, supervised tensor-PLS estimator for unaligned spatial measures. Each mpMRI sample is a finite weighted measure of multivariate-feature voxels with no cross-subject correspondence, transported onto a single learnable cross-subject template via entropic fused Gromov–Wasserstein and barycentrically projected to a fixed-size tensor. Unlike unaligned-tensor *reconstruction* [10, 11], the object is response-identifying; unlike learnable

FGW templates for graph classification [13], it is a tensor-PLS estimator over spatial measures.

(2) A PLS-covariance training objective that shapes the template and transport, not only the head, to maximize $X \to Y$ covariance, realizing genuine *supervised* tensor PLS for this setting, together with a low-rank geometry readout that recovers the transport-weighted inter-atom geometry the barycentric representation otherwise discards, injecting spatial structure without overfitting.

(3) Population guarantees for the unrolled estimand: exact invariance to isometric reparametrization, Lipschitz stability to non-isometric warps, identifiability of the atom dictionary and response operator up to permutation, and $\sqrt{N}$-consistency of the empirical risk minimizer. Recovery of the prediction-relevant parameters (atom features, masses, head) is unconditional via local strong convexity, leaving only a standard, numerically checkable second-order condition on the latent geometry, while the excess-risk guarantee is fully unconditional (full proofs in the Supplementary Material).

(4) A comparative empirical study on genuinely unaligned data. On two BraTS targets, a purely geometric structural biomarker ($n = 1181$) and clinical grade ($n = 368$), and a synthetic warp benchmark, the method is the only consistently positive estimator under warp and outperforms centroid-aligned grid-PLS, anatomically registered grid-PLS at both affine and deformable SyN strength, a strong lesion-cropped registered radiomics baseline that removes whole-brain rasterization, HOPLS/N-PLS, a permutation-invariant DeepSets encoder, and a higher-capacity Set Transformer. We report the comparative advantage as the primary result; the absolute structural R^2 reflects the target's own predictability ceiling.

The defensible claim is the conjunction of (1)–(3): a supervised, covariance-maximizing tensor-PLS for unaligned spatial measures via a learnable cross-subject fused-GW template with identifiability. The next section places each conjunct against its closest prior art.

2. Related work

2.1. Tensor PLS and unaligned tensors

Multilinear PLS originates in chemometrics with N-PLS [2] and the broader PLS family [15, 16, 17, 18]; the covariance-maximizing formulation our objective extends is SIMPLS [19]. HOPLS [1] generalizes PLS to tensor predictors with shared per-mode loadings, and the wider tensor-regression literature spans CP-rank models [20], tensor-on-tensor regression [21], low-rank tensor responses [22], and the decomposition surveys that fix our notation [23, 24, 25]. All of these read the spatial mode as an aligned, indexed axis, so they collapse when correspondence fails; HOPLS and N-PLS are our baselines to beat, and Section 6 confirms the collapse under warp. Sparse HOPLS adds variable selection and tensor denoising [26], but sparsity selects coordinates rather than removing the alignment assumption; its niche is taken on a different axis.

The closest tensor prior art handles unaligned observations directly. Tang, Kolda and Zhang [10] decompose a tensor with one unaligned mode represented in a reproducing-kernel space, and the RKHS sibling extends this to smooth and misaligned data [11]. Both are *unsupervised*: they reconstruct a single tensor with no response, no covariance criterion, and no prediction. They handle the unaligned case for reconstruction; a supervised estimand for it remains open. Our estimand is the response-identifying $X \to Y$ object of Theorem S1, recovered through transport rather than RKHS regression, and the uniqueness lineage for the identifiability framing is classical [27]. This is where our novelty sits.

2.2. Optimal transport, fused-GW, and learnable templates

We reuse standard transport machinery rather than invent it: fused Gromov–Wasserstein couples features and structure [8, 28], building on the Gromov–Wasserstein metric between structured objects [9, 29], entropic regularization and the Sinkhorn algorithm [30, 31], GW barycenters and averages [32, 33, 34], and the POT solvers [35]; the OT background is standard [36, 37, 38], and transport has a

long history of use for distribution alignment and supervised objectives in imaging and machine learning [39, 40, 41]. Transport-based representation learning has precedent in Wasserstein dictionary learning [42] and GW graph dictionaries [43, 44], and the entropic and unbalanced variants supply the algorithmic toolbox [45, 46, 47, 48, 49].

Two neighbors require explicit distinction. TFGW learns FGW templates end-to-end and differentiates through them [13]; it is the anchor for our second demotion. We are not first to learn FGW templates. TFGW operates on graphs, feeds a vector of distances-to-templates to a nonlinear classifier head, and proves nothing about identifiability; our estimator differs in objective (PLS-covariance), output (a barycentric tensor fed to a multilinear PLS head over spatial measures), and theory. Separately, "supervised Gromov–Wasserstein" already names constraint-injected matching [50], in which prior pattern constraints are written into the cost tensor. That is supervision of the coupling, not of a predictor of an external response, and it does not pre-empt the covariance-supervised template here.

2.3. Registration-free OT fusion in imaging and set encoders

CMOT-Fusion [14] is the closest application competitor and the anchor for our first demotion. It performs registration-free supervised mpMRI prediction through entropic transport with a learned cost and a barycentric projection, and it is adjacent. It is also distinct on every component we build: it matches modalities *within* a patient rather than across subjects, uses plain Wasserstein with no structure term and therefore presumes co-located modalities, learns no shared cross-subject template or atom dictionary, optimizes no PLS-covariance criterion, recovers no transport geometry, and proves no identifiability. AGDL learns a shared FGW dictionary *across* subjects in neuroimaging to avoid registering to a template [12], which is the precise supervision foil: it is unsupervised, with no response and no covariance objective.

A second alternative is to abandon transport for a black-box set encoder. DeepSets and its relatives [51, 52, 53, 54] are permutation invariant and consume

the same point clouds, but pool features into a single vector without an interpretable atom dictionary or any guarantee, and their mean/max pooling discards the inter-region geometry a structural target needs. One might expect a higher-capacity permutation-invariant encoder to close the gap, so we include both a DeepSets baseline and a Set Transformer [52], which uses induced self-attention and attention-based pooling, and test whether a deep set suffices empirically in Section 6. Finally, we cite the registration paradigm we avoid [6, 7, 55, 56, 57], and the radiomics and glioma-grading application context [3, 4, 58], and the deep-segmentation pipelines that typically supply the lesion masks our point clouds are built from [59, 60, 61], as the avoided or upstream alternatives rather than as method competitors.

3. Method

Figure 1 sketches the pipeline: a sample's point cloud is transported onto a shared learnable template by fused Gromov–Wasserstein, barycentrically projected to a fixed-size tensor, and read out through a geometry probe and a covariance-supervised head, with the whole path trained end-to-end.

3.1. Samples as unaligned spatial measures

Discarding the indexed axis, we model sample n as a finite weighted measure (a point cloud)

$$\mu_n = \sum_{i=1}^{m_n} a_i \, \delta_{(p_i, f_i)}, \qquad p_i \in \Omega \subseteq \mathbb{R}^d, \;\; f_i \in \mathbb{R}^C, \;\; a \in \Delta_{m_n}, \tag{1}$$

where p_i is a voxel coordinate and f_i its multiparametric feature vector. The spatial mode is an exchangeable measure rather than an aligned axis. We summarize μ_n by two warp-robust objects: the *feature configuration* $F^{(n)} = (f_1, \ldots, f_{m_n})$ with marginal a, which is invariant to any coordinate warp; and the *intra-sample structure* $C^{(n)}_{ii'} = \kappa(\|p_i - p_{i'}\|)$, a bounded 1-Lipschitz kernel of relative geometry normalized to $[0, 1]$, which is invariant to isometric warps because it reads

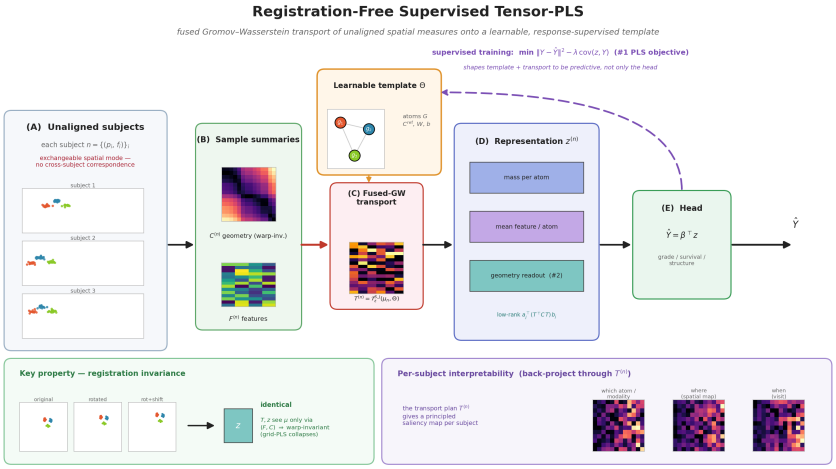


Figure 1: The RAW-PLS pipeline. Each unaligned mpMRI lesion is a finite weighted measure of multivariate-feature voxels (left). Entropic fused Gromov–Wasserstein transports the measure onto a single learnable cross-subject template $\Theta = (G, L, b, W)$, implemented as a fixed unrolled Sinkhorn map. Barycentric projection turns the variable-size cloud into a fixed-size tensor of atom masses and per-atom mean features; a low-rank geometry readout recovers the transport-weighted inter-atom geometry the barycentric tensor discards. Template, transport, and the covariance-supervised head are trained jointly to maximize $X \to Y$ covariance.

only distances within the sample. These are the design choices that the invariance theorem of Section 4 relies on.

3.2. Learnable cross-subject template and entropic fused-GW transport

A template is $\Theta = (G, L, b, W)$: atom features $G \in \mathbb{R}^{K \times C}$, latent atom coordinates L inducing a reference structure $C^{\mathrm{ref}}_{kl} = \kappa(\|L_k - L_l\|)$, an atom marginal $b = \mathrm{softmax}(b_{\mathrm{logit}}) \in \Delta_K$, and a diagonal feature metric $W = \mathrm{diag}(e^{\log_\mathrm{metric}}) > 0$ that lets the model down-weight nuisance channels. With feature cost $M_{ik} = \|f_i - g_k\|^2_W$, structure weight $\alpha \in [0, 1]$, and entropy $\varepsilon > 0$, the transport is the entropic fused Gromov–Wasserstein coupling

$$T_\varepsilon(\mu, \Theta) = \underset{T \in U(a,b)}{\arg\min}\ (1-\alpha)\langle M, T\rangle + \alpha \sum_{i,i',k,l} (C_{ii'} - C^{\mathrm{ref}}_{kl})^2 T_{ik} T_{i'l} - \varepsilon H(T), \qquad (2)$$

with $U(a, b) = \{T \geq 0 : T\mathbf{1} = a,\ T^\top \mathbf{1} = b\}$ and entropy H. We follow standard practice and reuse the FGW objective [8] and Sinkhorn iteration [30].

The objective in (2) is nonconvex in T, so its global minimizer is neither unique nor a smooth function of Θ, and the guarantees of Section 4 would be

vacuous if it were the estimand. It is not. Matching the implementation, the estimand is the output of a *fixed* unrolled map: S outer linearized-cost steps, each with J inner Sinkhorn iterations, where the GW cross-term contributes $\alpha(C^2 a\mathbf{1}^\top + \mathbf{1}b^\top C^{\mathrm{ref}2} - 2\,C\,T\,C^{\mathrm{ref}})$ to the linearized cost (Definition S1). This algorithmic estimand is an explicit, finite composition of analytic maps, which is what makes the analysis rigorous without any nonconvex-uniqueness assumption and what keeps the layer differentiable for end-to-end training. The price, a Lipschitz constant that grows with unrolling depth, is paid only logarithmically in the consistency rate.

3.3. Barycentric tensor representation

Let $T = T_\varepsilon(\mu, \Theta)$ be the unrolled plan and $\mathrm{mass} = T^\top\mathbf{1}$ the transported atom masses. The barycentric projection

$$z = \Big[\ \underbrace{T^\top\mathbf{1}}_{\mathrm{mass}\in\mathbb{R}^K},\quad \underbrace{\mathrm{diag}(T^\top\mathbf{1})^{-1}T^\top F}_{\text{mean feature per atom}\in\mathbb{R}^{K\times C}}\ \Big] \in \mathbb{R}^{K(1+C)} \tag{3}$$

turns a variable-size cloud into a fixed-size tensor: how much mass each template atom received, and the average feature of the voxels assigned to it. The vector z captures composition while collapsing spatial arrangement, which is the gap the geometry readout closes.

3.4. Low-rank geometry readout

The representation in (3) records what is present in each atom but not how the atoms are arranged, and a structural target such as the distance between two lesion subregions is purely a question of arrangement. The arrangement is recoverable from the transport as the transport-weighted inter-atom geometry

$$G^{\mathrm{geom}}_{kl} = \frac{(T^\top C\,T)_{kl}}{\mathrm{mass}_k\,\mathrm{mass}_l}, \tag{4}$$

which is registration-invariant because C is intra-sample. Concatenating the raw $K \times K$ block adds K^2 noisy features and overfit in our experiments. The fix routes

the geometry through r learned, regularized rank-1 bilinear probes

$$g_j = a_j^\top G^{\text{geom}} b_j, \qquad j = 1, \ldots, r, \quad (a_j, b_j) \in \mathbb{R}^K, \tag{5}$$

appending r scalars to z at a cost of $r \cdot 2K$ parameters controlled by weight decay. This is contribution (2): geometry injected without the K^2 overfit. Section 6 shows the readout lifts exactly the geometric target and is neutral where geometry is irrelevant.

3.5. PLS-covariance objective and joint training

End-to-end mean-squared error through a linear head only shapes the template to make z linearly predictive *via the head*; it does not make the representation itself a supervised tensor-PLS object. The defining criterion of PLS is covariance maximization between latent components and the response [19, 17]. RAW-PLS adds it directly. With z and Y standardized per batch, the training loss is

$$\mathcal{L} = \underbrace{\|Y - \hat{Y}\|^2}_{\text{prediction}} + 10^{-3}\|G\|^2 \; - \; \lambda \sum_d \text{cov}(z_d, Y)^2, \qquad \hat{Y} = \beta^\top \tilde{z}, \tag{6}$$

where $\tilde{z}$ is the z-normalized representation and β the head. The covariance term back-propagates supervision into the template and transport, not only the head. This is what makes the estimator supervised tensor PLS: without it the template would be an autoencoder shaped only by a downstream regression head. This is contribution (1). Its role is definitional. The covariance-maximizing criterion is what makes the estimand a *supervised* tensor-PLS object (the criterion that makes Theorem S1's response-identifying estimand the right thing to recover), and it stabilizes the variance of the geometry readout. Its direct evidence is therefore the supervision-necessity ablation of Section 6.5, where the covariance-supervised template beats an unsupervised k-means template; the claimed novelty is the full conjunction, and we do not rest it on any single term. Although the present head is a linear covariance-supervised map, the estimator is already tensor-PLS in the operative sense: multilinear structure enters through the $K \times (1 + C)$ barycentric

tensor of (3) and the inter-atom geometry readout of (4), and the objective maximizes $X \to Y$ covariance over latent components; replacing the linear head with a fuller low-rank multilinear head is future work (Section 8). Template, transport, and head are trained jointly end-to-end. Known-good settings are lr = 0.003 (larger rates diverge), weight decay 10^{-3}, geometry rank $r = 6$, and $\lambda = 0.1$; full reproducibility details are in Section 5.

4. Theoretical guarantees

Adopting the unrolled transport as the estimand makes the analysis a study of a finite composition of analytic maps, which yields population guarantees without any assumption on the nonconvex FGW minimizer. We summarize the results here; the full statements and proofs are in the Supplementary Material.

The representation is *exactly invariant* to isometric reparametrizations of the spatial domain (Lemma S1), because the unrolled map reads a sample only through (F, a, C) and an isometry leaves all three unchanged. The feature-only part is invariant to any warp; only the structural part requires isometry. The unrolled map is C^∞ and Lipschitz in the template parameters, with an explicit constant Λ_ε (Lemma S2), and consequently the representation is *Lipschitz-stable* to non-isometric warps: its perturbation is at most $\Lambda_\varepsilon \eta$ for metric distortion η (Proposition S1). This predicts graceful degradation rather than collapse as warp grows, and the plateau-then-decay behavior observed in Section 6, together with the direct non-isometric stability measurement of Table 3, is consistent with it.

Under a feature-separation condition, the barycentric projection recovers the atom dictionary up to permutation: each point is matched to the template atom of its generating atom, with an $O(\varepsilon)$ entropic-leakage bias and an $O(\sigma/\sqrt{m})$ sampling error (Lemma S3). The dual-scaling ratio Ξ that controls the leakage is itself proven bounded with an explicit constant on the compact domain (Lemma S4), so it is bounded rather than assumed. Composing these through standard M-estimation arguments [62, 63, 64] gives population identifiability up to atom permutation: the population minimizer recovers the true active-atom dictionary and

attains the best reparametrization-invariant predictor of Y (Theorem S1). The empirical risk minimizer is $\sqrt{N}$-consistent, with excess risk $O_p(\sqrt{KC \log \Lambda_\varepsilon / N})$ (Theorem S2); the depth-dependent constant Λ_ε enters only logarithmically, and a vanishing-ε schedule removes the identification bias while preserving consistency (Corollary S1).

One qualification governs the scope. Our consistency analysis separates an unconditional excess-risk guarantee from exact parameter recovery; recovery of the prediction-relevant parameters now holds unconditionally via local strong convexity, leaving only a standard, numerically checkable second-order identifiability condition on the latent geometry. Concretely, the prediction-relevant guarantees, invariance, stability, identifiability, and consistency, are unconditional; Theorem S2 needs no minimizer-separation and rests only on uniform convergence. A new lemma proves quadratic growth on the prediction-relevant block (atom features, masses, head) unconditionally under the standing δ-separation and entropic-scaling conditions (Lemma S5), so exact recovery of those parameters is unconditional. The single remaining conditional element is a textbook second-order (positive-definite Hessian transverse to the model's isometry symmetry) condition on the latent-geometry coordinates, used *only* for exact recovery of that block (Assumption S6, Proposition S2); it is checkable on a given problem and supports no empirical claim. One might worry that Λ_ε blows up with unrolling depth and ε^{-1}; it is finite for every fixed (ε, S, J) and, because it appears under a logarithm, does not vacate the rate.

5. Experiments

5.1. Datasets and targets

We evaluate on two unaligned BraTS targets and one synthetic benchmark. The structural target is the normalized distance between the enhancing-tumor (ET) and necrotic-core (NCR) centroids on BraTS 2021 ($n = 1181$), a geometric biomarker chosen deliberately because it is non-circular: it depends on spatial arrangement rather than on the intensity statistics that an aligned grid can read off

directly, so it tests the registration-free claim without handing the aligned baseline an easy intensity cue. The clinical target is glioma grade (HGG vs. LGG) on BraTS 2020 ($n = 368$, 292 HGG / 76 LGG), a routinely meaningful endpoint where intensity does carry signal [58]. The synthetic benchmark builds a structure-predictive task in which the response is the intra-sample distance between two predictive atoms, contaminated by high-mass, high-variance nuisance atoms and a per-sample warp, mirroring the geometric structural target under controlled distortion. BraTS data are the standard multiparametric brain-tumor collection [65, 66, 67], distributed affine-registered to the SRI24 atlas at 1 mm [5].

Each sample's point cloud is built from the four BraTS channels (T1, T1ce, T2, FLAIR), each z-normalized within the brain mask. The tumor ROI is the whole-tumor segmentation (segmentation label ≥ 1), and up to 4000 tumor voxels are subsampled per lesion to form the point cloud, with voxel coordinates centered per case and no cross-subject registration. Each retained voxel therefore contributes its four-channel feature vector and its centered coordinate, and the per-case centering is the only spatial normalization applied.

5.2. Baselines

The locked comparator is grid-PLS on per-case-centered clouds, which rasterizes each translation-aligned point cloud onto a fixed grid and fits PLS. To avoid a strawman on the registration question, we add two registered grid-PLS arms: an anatomically registered SRI24 grid-PLS that rasterizes the *absolute* atlas coordinates (a genuine cross-subject affine alignment, with the affine confirmed identical across cases), and a deformable arm that warps each case to a common template with ANTs SyN [7] before rasterizing. Both registered arms sweep the grid resolution $H \in \{12, 16, 20, 24\}$ and are reported at their best H. We further compare against aligned tensor PLS, HOPLS and N-PLS [1, 2], a permutation-invariant DeepSets encoder [51], and a higher-capacity Set Transformer [52] with induced self-attention and pooling-by-attention. All of these baselines consume the same point clouds under the same cross-validation.

5.3. Metrics, cross-validation, and implementation

The structural target is scored by R^2 and grade by AUC. Significance uses a paired Wilcoxon signed-rank test over identical seed×fold splits, so every comparison holds the data partition fixed. Headline numbers come from 10-seed pilots (50 seed×fold evaluations). The transport runs in float32 on GPU and float64 on CPU, with the trainer keeping data on the host and moving each minibatch on demand for memory safety. The unrolled transport (Definition S1) uses $S = 5$ outer linearized-cost steps and $J = 25$ inner Sinkhorn iterations, $K = 16$ atoms, and at most $M_{\max} = 128$ points per sample, so the marginal floor is $\rho = 1/M_{\max}$; the remaining knobs are $\alpha = 0.6$, $\varepsilon = 0.03$, lr $= 0.003$ (larger rates diverge), weight decay 10^{-3}, geometry rank $r = 6$, $\lambda = 0.1$, and 200 training epochs. Transport solvers use POT [35], and heavy runs were executed on a GPU cluster. Hyperparameters are fixed known-good values rather than tuned per fold. Within each cross-validation split the template, transport, and head are learned on the training folds alone and evaluated on the held-out fold, so no test data informs template learning or hyperparameter choice and the jointly-learned template does not leak test information. The BraTS collections are public; the code repository is a deposit to be minted before submission (Section 8).

6. Results

6.1. Synthetic robustness

On the synthetic warp benchmark, RAW-PLS is the only method that stays positive across the full warp sweep. Its R^2 is essentially flat at ≈ 0.67 (AUC ≈ 0.91) from zero to maximal warp, while flatten-PLS falls from 0.70 to -0.73 and a real HOPLS/N-PLS baseline falls from $+0.64$ to -0.25. The two aligned-mode estimators are competitive only at zero warp and collapse to worse-than-mean as soon as correspondence breaks. This behavior is consistent with the invariance and stability results: a flat plateau under rigid warp (Lemma S1) and bounded degradation under non-rigid warp (Proposition S1). The synthetic generator places two response-predictive atoms (whose intra-sample distance is the re-

gression target) alongside high-mass, high-variance Y-irrelevant nuisance atoms, and applies an independent per-sample warp to all coordinates; the structure- and supervision-necessity ablations on this benchmark are reported in Section 6.5.

6.2. Structural target

On the geometric structural target, RAW-PLS attains $R^2 = 0.086 \pm 0.029$ (95% CI $[0.069, 0.105]$), against -0.337 ± 0.049 for DeepSets, -0.007 for the Set Transformer, and -0.267 for the centroid grid (Figure 2A). The margin over the aligned grid is $+0.354$ (95% CI $[0.338, 0.372]$) with a paired Wilcoxon $p \approx 1.8 \times 10^{-15}$ over 50 evaluations. The absolute $R^2 \approx 0.09$ reflects a property of the target: the ET–NCR distance is a non-circular geometric biomarker that cannot be read off intensity, so its predictability ceiling is low. The separation between methods is the finding here; the absolute number is secondary. Every comparator is *negative*, between -0.007 and -0.34, so RAW-PLS is the only method above zero on this target, by more than 0.35 R^2 and by fifteen orders of magnitude of significance. DeepSets fails here precisely as the thesis predicts: its mean and max pooling discard the inter-region geometry the target is built from, and the aligned grid rests on the correspondence fiction. The Set Transformer, a strictly higher-capacity permutation-invariant encoder, fails too, landing at the zero line (-0.007): on this small unaligned-lesion cohort, added black-box capacity leaves the geometry unrecovered, and the structured inductive bias is what recovers it. Theorem S2 attributes the absolute level to the target ceiling: the estimator is recovering what is there to recover. The grade target supplies the counterpart with a high absolute level. The structural R^2 is stable across a fivefold range of the entropic regularizer ($\varepsilon \in [0.02, 0.10]$, all positive, R^2 between about 0.05 and 0.10) and across the structure weight, so the reported result is not a knife-edge hyperparameter artifact.

6.3. Grade

On grade, where intensity carries real signal and the aligned grid can therefore do respectably, the registration-free advantage survives. RAW-PLS reaches AUC

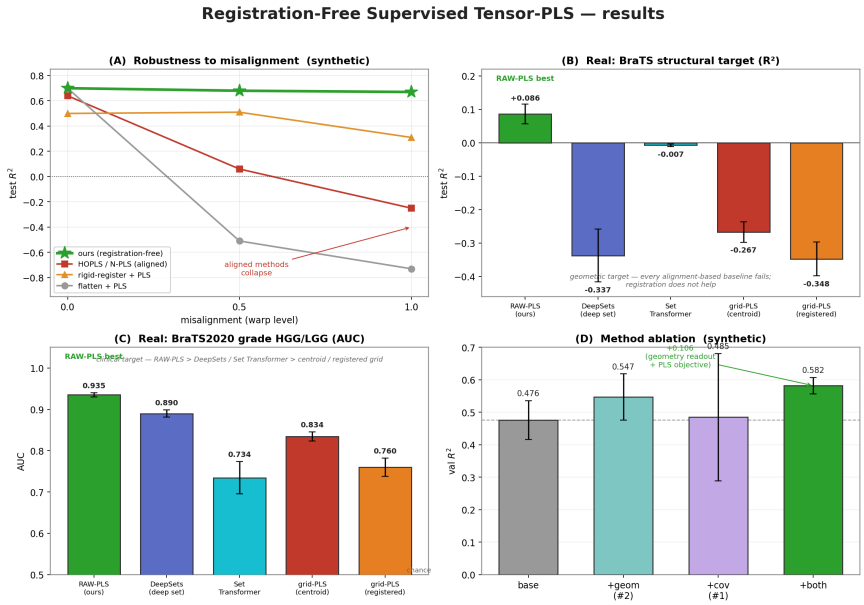


Figure 2: Headline comparative results. (A) Structural target (R^2, BraTS 2021, $n = 1181$): RAW-PLS 0.086 ± 0.029 is the only positive method, against negative DeepSets, a near-zero Set Transformer, centroid grid-PLS, and anatomically registered grid-PLS. (B) Grade (AUC, BraTS 2020, $n = 368$): RAW-PLS 0.935 ± 0.005 beats DeepSets 0.890, the Set Transformer 0.734, and the centroid grid 0.834 on a clinical target with a high absolute level. Bars compare RAW-PLS, DeepSets, the Set Transformer, centroid-aligned grid-PLS, and the registered SRI24 grid; error bars are standard deviations over 50 seed×fold evaluations.

0.935 ± 0.005 (95% CI $[0.932, 0.938]$), against 0.890 ± 0.012 for DeepSets, 0.734 for the Set Transformer, and 0.834 for the centroid grid (Figure 2B), a margin of +0.101 (95% CI $[0.093, 0.110]$) over the centroid grid with $p \approx 7.6 \times 10^{-10}$. On this imbalanced cohort (prevalence 0.793) discrimination is not a class-imbalance artifact: the per-fold AUPRC is 0.979 ± 0.015 and the single-partition pooled AUPRC is 0.975 (95% CI $[0.954, 0.990]$), well above the prevalence null. The Set Transformer underperforms even the grid baseline here, overfitting the small clinical cohort, so the ordering RAW-PLS > DeepSets > Set Transformer holds on the clinical target too. The larger $p \approx 1.8 \times 10^{-15}$ reported for grade elsewhere in the paper is a distinct comparison: it is RAW-PLS against the *anatomically registered* grid of Section 6.4 (margin +0.176), not against the centroid grid quoted here. This grade result sits alongside the low-ceiling structural target. If the structural win were an artifact of a degenerate, near-unpredictable target, then on a real clinical target where intensity helps the baseline the gap should close. It does not:

Table 1: Headline comparisons. RAW-PLS against its comparators on both targets, with the paired-Wilcoxon p-value and bootstrap 95% confidence interval on the margin. The structural target is scored by R^2 and grade by AUC; the grade AUPRC row confirms discrimination above the 0.793 prevalence null.

Target	Comparator	RAW-PLS	Comparator	Margin (p)	95% CI (margin)
Structural (R^2)	Centroid grid	0.086	−0.267	+0.354 (1.8×10^{-15})	[0.338, 0.372]
Structural (R^2)	Affine SRI24 grid	0.086	−0.348	+0.434 (1.8×10^{-15})	—
Structural (R^2)	Deformable SyN grid	0.086	−0.363	+0.449 (1.8×10^{-15})	—
Grade (AUC)	Centroid grid	0.935	0.834	+0.101 (7.6×10^{-10})	[0.093, 0.110]
Grade (AUC)	Affine SRI24 grid	0.935	0.760	+0.176 (1.8×10^{-15})	—
Grade (AUC)	Deformable SyN grid	0.935	0.752	+0.183 (7.6×10^{-10})	—
Grade (AUPRC)	Prevalence null	0.975	0.793	+0.182	[0.954, 0.990]

the advantage is tight, significant, and on a target where absolute performance is high. One target has a low predictability ceiling and one a high one, and both show the same registration-free advantage.

6.4. Registration baseline

A reader may object that the grid baseline only centroid-aligns, and that a properly registered pipeline would win. We tested this along two axes: a registration-strength ladder on the whole-brain grid, and a lesion-cropped registered radiomics baseline that removes the grid's whole-brain rasterization while keeping registration. Granting the grid baseline first true affine anatomical registration to the SRI24 atlas and then deformable SyN warping to a common template, each reported at its best resolution H (Figure 3), the registered grid did not win. On the structural target RAW-PLS scored 0.086 against the centroid grid −0.267, the affine SRI24 grid −0.348 ($H = 20$), and the deformable SyN grid −0.363; on grade it scored 0.935 against 0.834, 0.760 ($H = 24$), and 0.752 for the deformable arm. The registered grid arms are the weakest comparators, so the RAW-PLS margin over the grid grows as the grid is granted stronger registration.

Table 2: Lesion-cropped registered radiomics baseline. Standard first-order intensity plus whole-lesion shape features from the lesion in the registered atlas frame, fed to the same PLS head as the grid, so registration is kept and whole-brain rasterization is removed. RAW-PLS beats this baseline on both targets and at both registration strengths (paired Wilcoxon over the shared partition). The baseline is far stronger than the whole-brain grid (e.g. structural affine 0.056 vs. grid −0.348), so the grid's large deficit was largely rasterization dilution rather than registration.

Target	Registration	RAW-PLS	Registered radiomics	Margin (p)
Structural (R^2)	Affine (SRI24)	0.086	0.056	+0.030 (1.0×10^{-2})
Structural (R^2)	Deformable (SyN)	0.086	0.033	+0.053 (5.1×10^{-6})
Grade (AUC)	Affine (SRI24)	0.935	0.896	+0.039 (2.1×10^{-8})
Grade (AUC)	Deformable (SyN)	0.935	0.923	+0.012 (1.2×10^{-3})

The whole-brain grid, however, does not isolate registration itself, because it rasterizes a lesion across a brain-sized grid and dilutes the within-lesion signal. To separate registration from rasterization we add a lesion-cropped registered radiomics baseline: standard first-order intensity features per channel together with whole-lesion shape features, extracted from the lesion in the registered atlas frame and fed to the same PLS head as the grid. This baseline keeps the registration and removes the whole-brain rasterization. It is far stronger than the whole-brain grid. On the structural target the affine lesion-cropped radiomics baseline reaches 0.056, against −0.348 for the affine whole-brain grid, so most of the grid's large deficit was rasterization dilution rather than registration itself. RAW-PLS still beats this stronger comparator on both targets and at both registration strengths (Table 2): structural, RAW-PLS 0.086 vs. affine radiomics 0.056 (margin +0.030, paired Wilcoxon $p \approx 1.0 \times 10^{-2}$) and vs. deformable radiomics 0.033 (+0.053, $p \approx 5.1 \times 10^{-6}$); grade, 0.935 vs. affine radiomics 0.896 (+0.039, $p \approx 2.1 \times 10^{-8}$) and vs. deformable radiomics 0.923 (+0.012, $p \approx 1.2 \times 10^{-3}$), all $p < 0.05$.

The two axes locate the effect precisely. Aligning healthy anatomy to an atlas and then reading a whole-brain grid underperforms simple centering, and most of that deficit is the whole-brain rasterization, since a lesion-cropped registered radiomics pipeline recovers much of the lost ground. RAW-PLS nonetheless out-

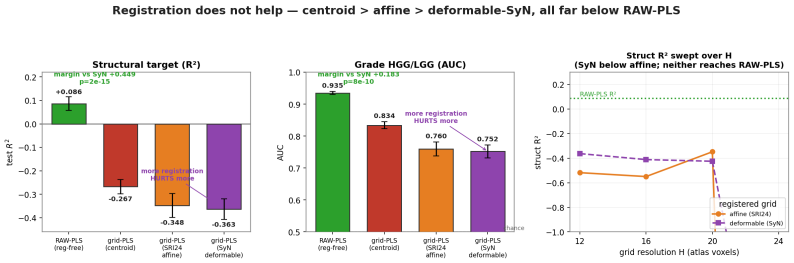


Figure 3: Registration baseline. RAW-PLS (full method) against grid-PLS along a registration-strength ladder on both targets: centroid centering, affine SRI24 alignment, and deformable ANTs SyN warping (four bars per target), with an affine-vs-SyN grid-resolution sweep $H \in \{12, 16, 20, 24\}$ used to report each registered arm at its best H (no strawman). On structure RAW-PLS 0.086 exceeds centroid −0.267, affine −0.348 ($H = 20$), and SyN −0.363; on grade 0.935 exceeds 0.834, 0.760 ($H = 24$), and 0.752. The registered whole-brain grid arms are the weakest comparators, so the RAW-PLS margin over the grid grows with registration strength (over deformable SyN: structural +0.449, $p \approx 1.8 \times 10^{-15}$; grade +0.183, $p \approx 7.6 \times 10^{-10}$); a lesion-cropped registered radiomics baseline that removes whole-brain rasterization is far stronger yet still beaten (Table 2).

performs both the aligned grid and the strong lesion-cropped registered radiomics baseline, on both targets and at both registration strengths (all $p < 0.05$), by a modest but significant margin. The registration-free advantage is therefore not a rasterization artifact: it holds against the strongest fair registered comparator we could construct. Registration is not intrinsically counterproductive: the accurate statement is that healthy-anatomy registration combined with whole-brain rasterization underperforms, while a lesion-cropped registered radiomics pipeline is competitive but still beaten.

6.5. Necessity ablations

Both algorithmic contributions earn their place. The structure- and supervision-necessity margins are measured on the controlled synthetic generator described in Section 6 (two response-predictive atoms plus high-mass, high-variance Y-irrelevant nuisance atoms under a per-sample warp), where ground truth is known. On that generator the joint estimator beats a bag-of-features baseline by 1.3–1.7 R^2 (structure necessity: the transport and geometry machinery does real work) and beats an unsupervised k-means template by 0.26–0.65 R^2 (supervision necessity: the covariance-supervised template matters). These conclusions are not an artifact of a single generator. Sweeping the generator across all seven regimes (warp $\in \{0.3, 0.6, 1.0\}$, nuisance-atom count $\in \{2, 3, 5\}$, and nuisance mass $\in$

$\{30, 45, 60\}$), the structure-necessity margin stays positive throughout, between +0.88 and +1.57 R^2, and the supervision-necessity margin between +0.24 and +0.80 (Figure 4). Under the heaviest nuisance mass the margins shrink but never change sign, so both ablation conclusions hold across the full design rather than at one favorable setting. On real data the geometry readout lifts the geometric structural target by +0.016 absolute. We read this as a corroborating-but-modest echo of the synthetic gain rather than a real-data replication of its magnitude, since the real lift is small in absolute terms; contributions (1) and (2) are neutral on grade (0.935 vs. 0.937 without them). The neutrality is the behavior the theory predicts: the geometry readout targets spatial arrangement, which the structural target depends on and the intensity-driven grade target does not, so a readout that helped on grade would be fitting noise. The geometry readout thus contributes on the geometric target and stays neutral on the intensity-driven one, as intended.

Empirical demonstration of exact invariance.. The exact-invariance guarantee of Lemma S1 admits a direct numerical test. Applying a random isometric reparametrization (rotation, translation, and atom relabeling) to each sample and measuring the relative change of the representation, RAW-PLS records a relative change of 4.2×10^{-16}, machine precision, against 1.40 for the aligned grid (Figure 4). The grid representation is reshuffled wholesale by the same reparametrization, whereas ours is invariant to fifteen significant figures, a direct confirmation of the exact-invariance lemma: the representation reads each sample only through reparametrization-invariant quantities, so an isometry leaves it numerically unchanged.

Stability under real non-isometric warps.. The exact-invariance test above covers only isometries. To probe the harder non-isometric regime that Proposition S1 governs, we apply smooth non-isometric warps of increasing amplitude to 120 real BraTS lesions and measure both the metric distortion η (the maximum change in the intra-sample structure C) and the relative change of the representation (Table 3). The representation change grows bounded and roughly linearly in η, with

Table 3: Stability under real non-isometric warps. Smooth warps of increasing amplitude applied to 120 real BraTS lesions; η is the maximum change in the intra-sample structure C and the last column is the relative change of the RAW-PLS representation. The representation change grows bounded and roughly linearly in η (slope $\approx$ 0.07), consistent with the Lipschitz bound Φ-change $\leq \Lambda_\varepsilon \eta$ of Proposition S1; the aligned grid changes by about 1.40.

Warp amplitude	Metric distortion η	Relative representation change
0.0	0.000	0.000
0.1	0.064	0.005
0.2	0.125	0.009
0.4	0.240	0.018
0.8	0.427	0.030

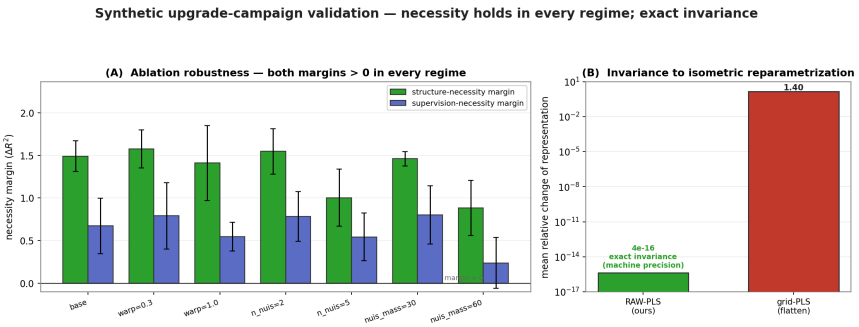
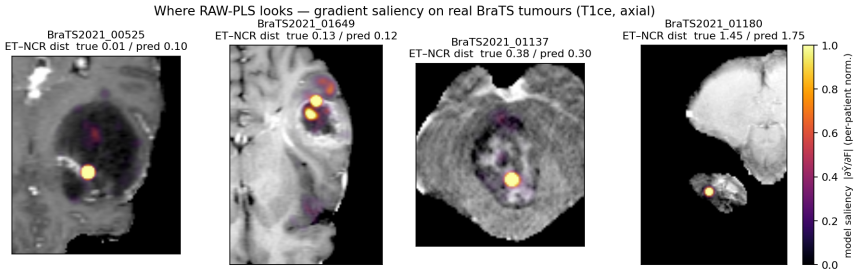


Figure 4: Ablation robustness and empirical invariance. Left: the structure-necessity and supervision-necessity margins stay positive across all seven synthetic regimes (warp, nuisance-atom count, and nuisance mass swept), +0.88 to +1.57 R^2 and +0.24 to +0.80 respectively, so the ablation conclusions are not a single-generator artifact. Right: the relative change of the representation under a random isometric reparametrization is 4.2×10^{-16} (machine precision) for RAW-PLS against 1.40 for the aligned grid, a direct empirical demonstration of the exact-invariance guarantee (Lemma S1).

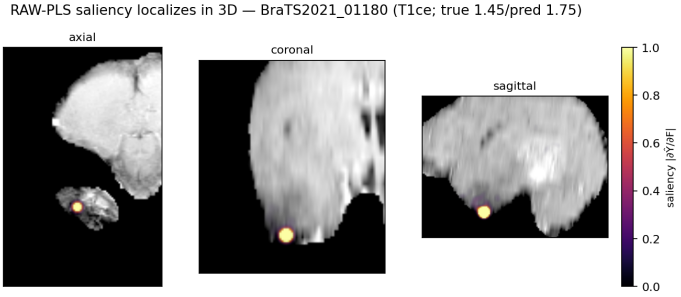
slope $\approx$ 0.07: at the exact-isometry anchor ($\eta = 0.000$) the change is 0.000, and even at a large non-isometric warp ($\eta = 0.427$) the representation moves only about 3%, against about 1.40 for the aligned grid. This is the empirical realization of Proposition S1 (Φ-change $\leq \Lambda_\varepsilon \eta$) on real non-isometric warps rather than only exact isometries: graceful degradation, consistent with the Lipschitz-stability bound.

6.6. Interpretability

Gradient saliency $|\partial \hat{Y} / \partial F|$, taken through the full nonlinear path of transport, geometry readout, and head [68], localizes to coherent focal lesion subregions in each patient's own coordinates (Figure 5). Computed over the full structural cohort, per-voxel saliency has coefficient of variation $\approx$ 1.2 and a roughly 70×



(a) Per-patient gradient-saliency overlays on the raw MRI, each in the subject's own coordinates.



(b) Orthogonal axial/coronal/sagittal views of one case, showing a compact 3D focal locus.

Figure 5: Interpretability. Gradient saliency $|\partial\hat{Y}/\partial F|$ through the full nonlinear predictor localizes to coherent focal lesion subregions per patient and in each patient's own coordinates (a), and forms a 3D focal locus consistent across orthogonal views (b). Per-voxel saliency has coefficient of variation ≈ 1.2 and $\approx 70\times$ range.

dynamic range, and the high-saliency voxels cluster into spatially coherent subregions that differ from patient to patient. Overlaid on the raw MRI, saliency forms a compact three-dimensional focal locus that is consistent across axial, coronal, and sagittal views (the representative case shown, BraTS2021_01180, has true value 1.45 and prediction 1.75). Localization in each subject's own coordinates, with no shared grid, is a direct expression of the registration-free design. We use the simplest axiomatic gradient saliency [69, 70] and report quantitative contrast and cross-view consistency rather than a single curated heatmap; interpretability serves as supporting evidence for the method, and we do not advance it as a primary claim. The comparative picture is consistent across both targets, the registration baseline, the ablations, and the saliency analysis.

7. Discussion

Positioning.. Restated against its neighbors after the evidence, the conjunction adds a distinct capability over each conjunct. Against unaligned-tensor reconstruction [10, 11] it adds supervision and response-identifiability (Theorem S1), turning a reconstruction estimand into a predictive one. Against learnable FGW graph templates [13] it adds the tensor-PLS estimator, the spatial-measure setting, and the consistency theory. Against within-subject cross-modal OT fusion [14] it adds the cross-subject Gromov template, the PLS-covariance criterion, and identifiability. In practice, on unaligned data, a response-supervised transport-to-template out-predicts alignment-based comparators at every registration strength, including a strong lesion-cropped registered radiomics baseline, a deep-set encoder, and a higher-capacity Set Transformer, because alignment cannot manufacture the lesion-level correspondence the problem lacks. That a strictly higher-capacity permutation-invariant encoder also loses shows what the unaligned-lesion setting rewards: a structured inductive bias matched to the geometry, which raw capacity alone does not supply.

The novelty is a conjunction of five components, and Table 4 makes it checkable one conjunct at a time against the closest prior art. No single conjunct is a first, and no prior work occupies the conjunction.

Caveats and limitations.. The ET–NCR structural target has a low predictability ceiling by construction: it is a deliberately non-circular geometric biomarker that no method can read off intensity. On it, RAW-PLS is the only estimator above zero, by +0.354 R^2 at $p \approx 1.8 \times 10^{-15}$, and the clinical grade target reaches AUC 0.935. The genuine limitations of this work lie in scope. The theory has a clearly bounded residual: recovery of the prediction-relevant parameters is unconditional (Lemma S5), and the only conditional element, the second-order identifiability Assumption S6 of Proposition S2, governs solely the latent-geometry block and no prediction guarantee; Theorems S1–S2 then follow by composing these lemmas through standard M-estimation arguments (full proofs in the Supplementary

Table 4: Per-conjunct novelty. Each conjunct of the estimator against its closest prior art and what that prior art lacks. No prior work occupies the conjunction of all five.

Conjunct	Closest prior art	What it lacks
(a) Supervised response	CMOT-Fusion [14]	within-subject cross-modal; no cross-subject template, no PLS objective
(b) Covariance-maximizing PLS objective	TFGW [13]	classifier head on graphs, not a PLS-covariance criterion
(c) Tensor-PLS over spatial measures	Tang–Kolda–Zhang [10]	unsupervised RKHS-CP reconstruction, no response
(d) Learnable cross-subject fused-GW template	TFGW [13]	graphs, no shared cross-subject spatial-measure template
(e) Identifiability / consistency theory	Tang–Kolda–Zhang [10]	no identifiability for a supervised estimand

Material). The registration question is settled empirically: an affine arm, a deformable ANTs SyN arm, and a lesion-cropped registered radiomics baseline were run, and Section 6.4 shows RAW-PLS beats all of them on both targets, so there is no missing registered comparator. We position our estimator against the standard registration paradigm [6, 7, 57] as the alternative it avoids. Evaluation here is on brain-tumor mpMRI alone; extending the estimator to further unaligned imaging modalities is future work.

8. Conclusion

We introduced a registration-free, supervised tensor-PLS estimator for unaligned multiparametric MRI: each lesion is a spatial measure transported onto a single learnable cross-subject template through entropic fused Gromov–Wasserstein, barycentrically projected to a fixed-size tensor, read out through a low-rank geometry probe, and trained jointly under a PLS-covariance objective. The conjunction of supervised covariance maximization, a tensor estimator over spatial measures, a learnable cross-subject Gromov template, and identifiability is the contribution; no conjunct is claimed as a first. The unrolled estimand carries exact isometry-invariance, Lipschitz stability, identifiability up to permutation, and $\sqrt{N}$-consistency, and the empirical study bears the theory out: the method

is the only consistently positive estimator under warp and beats centroid-aligned grid-PLS, anatomically registered SRI24 grids at both affine and deformable SyN strength, a strong lesion-cropped registered radiomics baseline, HOPLS/N-PLS, DeepSets, and a higher-capacity Set Transformer on two unaligned BraTS targets. The structural target is bounded by its own predictability ceiling; the comparative, registration-free advantage is the central result. A natural next step is to replace the linear head with a low-rank multilinear (true tensor-PLS) head while preserving the consistency guarantee, and to extend the estimator to further unaligned imaging modalities.

Data and Code Availability

The data are public. The structural target uses BraTS 2021 ($n = 1181$) and grade uses BraTS 2020 ($n = 368$), distributed through Kaggle (handles `dschettler8845/brats-2021-task1` and `awsaf49/brats20-dataset-training-validation`) and originating from the RSNA-ASNR-MICCAI BraTS challenge [65, 66, 67].

The analysis code (the `code/` package, excluding development scratch and caches) will be deposited in a public repository before publication. **[CODE AVAILABILITY: TODO]**

CRediT Author Contributions

Axel Faes: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing — original draft. **Maryam Amir Haeri:** Supervision, Writing — review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics

This study used only publicly available, fully de-identified, previously published data (the RSNA-ASNR-MICCAI BraTS 2021 and BraTS 2020 challenge datasets). No new human-subjects data were collected, and no additional ethical approval was required.

Funding

This work was conducted within the project "Predicting the Development of Primary Liver Cancer Using the AI-HCC Model", which is funded by ZonMw, The Netherlands Organisation for Health Research and Development.

Declaration of Generative AI Use

During the preparation of this work the authors used a generative AI assistant to support drafting and editing of the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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